

2.1. $B = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right)$ hledáme vektor $\langle (x_1, x_2), (y_1, y_2) \rangle$ v mřížce
 m x_1, x_2, y_1, y_2

TVZ 8.50

$$\langle u, v \rangle = [u]_B^* [v]_B$$

$$\langle b_1, b_2 \rangle = 1 \quad \langle (1, 2), (1, 2) \rangle = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1$$

$$\langle (1, 2), (1, 2) \rangle = a \cdot 1 \cdot 1 + b(2 \cdot 1 + 1 \cdot 2) + c \cdot 2 \cdot 2 = a + 4b + 4c = 1$$

$$\langle (3, 7), (3, 7) \rangle = 9a + 42b + 49c = 1$$

$$\langle (1, 2), (3, 7) \rangle = 3a + 13b + 14c = 0$$

Nejdříve uvažujeme, že ON báze má normu 1 a poté, se ukáže, že součin báze je nulový

dokazujeme:

$$a + 4b + 4c = 1$$

$$9a + 42b + 49c = 1$$

$$3a + 13b + 14c = 0$$

$$\left(\begin{array}{ccc|c} 1 & 4 & 4 & 1 \\ 9 & 42 & 49 & 1 \\ 3 & 13 & 14 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 4 & 4 & 1 \\ 0 & 6 & 13 & -8 \\ 0 & 1 & 2 & -3 \end{array} \right) \sim$$

$$\left(\begin{array}{ccc|c} 1 & 4 & 4 & 1 \\ 0 & 6 & 13 & -8 \\ 0 & 1 & 2 & -3 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 4 & 4 & 1 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 1 & 10 \end{array} \right) \quad \begin{array}{l} c = 10 \\ b = -23 \\ a = 53 \end{array}$$

$$\langle (x_1, x_2), (y_1, y_2) \rangle = 53x_1y_1 - 23x_2y_1 - 23x_1y_2 + 10x_2y_2$$